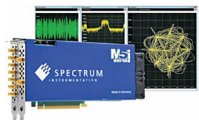


nal processing and analysis. Both PCIe digitizers deliver optimal GHz signal acquisition and analysis capabilities. The one- and two-channel cards offer a stunning combination of ultrafast 10GS/s sampling speed with 12-bit vertical resolution and 12.8GB/s data



streaming (over the PCIe bus). The cards feature front-end circuitry with over 3GHz bandwidth and up to 16GB (8GSamples) of on-board memory. The PCIe digitizer cards are ideal for users who need to perform data intensive processing, like continuous averaging for noise reduction, or MPoint-FFTs for spectral analysis.

Spectrum Instrumentation
<https://spectrum-instrumentation.com>

RF power measurement sensors

R&S NRP90S(N) from Rohde & Schwarz are diode power sensors for power measurements up to 90GHz frequency. The new instruments offer a frequency range of 50MHz to 90GHz with a dynamic range from -70dBm to 20dBm along with high measurement speeds of 50,000 per second. The



higher measurement frequency of sensors allows the instruments to cover the latest spectrum allocation for 5G FR2-2 up to 71GHz, satellite communications in both the 71GHz to 76GHz band and the 81GHz to 86GHz band, automotive radars operating from 76GHz to 81GHz, as well as other lower frequency wireless transmission technologies. The NRPxxS power sensors have a portable form factor and flexible operating modes that makes them suitable for both local and remote installation in maintenance or monitoring applications. It also enables high-speed power measurements for all frequencies above 67GHz in 5G, automotive radar, and satellite communications applications,

including the entire E band.

Rohde & Schwarz

<https://www.rohde-schwarz.com>

MISCELLANEOUS

Edge-AI developer cloud

STMicroelectronics has unveiled the world's first microcontroller (MCU) edge-AI developer cloud. The STM32Cube.AI developer cloud opens access to an extensive suite of online development tools built around the STM32 family of MCUs. The STM32Cube.



AI desktop front-end includes the resources for developers to validate and generate optimised STM32 AI libraries from trained neural networks. The STM32Cube.AI, which is an online tool, offers interface to generate optimised C-code for STM32 microcontrollers, without requiring prior software installation. This new tool brings the possibility to remotely benchmark models on STM32 hardware through the cloud to save on workload and cost. It opens access to the STM32 model zoo, a repository of trainable deep-learning models and demos to speed application development. The tool can be used for various use cases, including human motion sensing for activity recognition and tracking, computer vision for image classification or object detection, and audio event detection for audio classification.

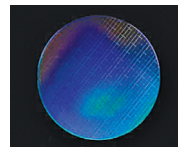
STMicroelectronics

<https://www.st.com>

Quantum software development kit

Intel has released their quantum software development kit (SDK) version 1.0 to grow the developer ecosystem. The SDK is a full quantum computer in simulation that can also interface with Intel's quantum hardware, including Intel's Horse Ridge II control chip and the upcoming Intel's quantum spin qubit chip. The kit allows

developers to program quantum algorithms in simulation, and it features an intuitive programming interface written in C++ using an industry-standard low-level virtual machine (LLVM)



compiler toolchain.

As a result, Intel's SDK offers seamless interfacing with C/C++ and Python applications, making it more versatile and customisable. The Intel Quantum SDK helps programmers get ready for future large-scale commercial quantum computers. It will help developers learn how to create quantum algorithms and applications in simulation.

Intel

<https://www.intel.com>

Printed electronic kit for IoT

Henkel INKxperience is a novel sensor experience kit tailor-made for IoT engineering across industries. The novel Henkel Qhesive Solutions Sensor INKxperience kit offers four different printed electronics technologies that are



pre-configured with hardware and software for prototyping and engineering ideation. The kit has been designed for professional engineers to help them quickly and efficiently explore and experiment with the technology of printed electronics for the development of IoT sensor solutions. The kit contains four different technologies, including a leak detection sensor, a non-contact liquid level sensor, a positive temperature coefficient heater, and a single- as well as multi-zone force sensitive resistor sensor, which are all reusable multiple times. The printed technologies are pre-configured with hardware and software for immediate and continuous data readout and offer an easy setup for the ideation of new ideas and proof of concept during prototyping and the engineering process of next-generation IoT sensors.

Henkel

<https://www.henkel-adhesives.com>